

Linux & Git Cheat Sheet

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ITEC 111 — Online Activity #2

01 Basic Linux Commands

pwd	Print the current working directory path
ls -la	List all files and folders in a directory
cd folder/	Change or go to the specific folder
mkdir project	Create a new directory named "project"
rm -r project/	Remove a directory and its contents
cp file.txt bak.txt	Copy a file to a new file name or location
mv old.txt new.txt	Move or rename a file
cat file.txt	Print the full contents of a file
nano file.txt	Open a file in the Nano text editor
chmod +x script.sh	Make a file executable
grep "text" file.txt	Search a file for lines containing "text"
sudo apt update	Refresh the list of available packages

02 Basic Git Commands

git init

Turns the current folder into a new, empty Git repository.

Ex: git init **Flags:** --bare

git status

Shows which files are changed, staged, or untracked right now.

Ex: git status **Flags:** -s (short)

git add

Moves changes from the working directory into the staging area.

Ex: git add index.html **Flags:** -A , .

git commit

Saves the staged changes as a permanent snapshot in history.

Ex: git commit -m "msg" **Flags:** -m, -am

git log

Lists past commits, newest first, with author and message.

Ex: git log --oneline **Flags:** --oneline, --graph

git diff

Shows the line-by-line difference between file versions.

Ex: git diff **Flags:** --staged

git restore

Discards local changes or unstages a file back to the last commit.

Ex: git restore file.txt **Flags:** --staged

git branch

Lists, creates, or deletes branches in the repository.

Ex: git branch feature **Flags:** -d, -a

git checkout / switch

Switches the active branch, or restores files from another commit.

Ex: git switch main **Flags:** -b (new branch)

git merge

Combines the history of another branch into the current branch.

Ex: git merge feature **Flags:** --no-ff

git tag

Marks a specific commit, usually to flag a release version.

Ex: git tag v1.0 **Flags:** -a, -d

git clone

Downloads a full copy of a remote repository onto your machine.

Ex: git clone <url> **Flags:** --depth

03 Git's Three-Tree Architecture

1. Working Directory

Your actual project folder on disk. Where you edit files directly. Git sees these edits as "unstaged changes" until you tell it otherwise.

**git
add**
→

2. Staging Area (Index)

A holding area for changes you've marked with git add. Lets you build a commit out of only the edits you choose, not everything at once.

**git
commit**
→

3. Repository (.git)

The permanent, saved history. Once git commit runs, staged changes become a snapshot here that can be viewed, compared, or reverted to.

Reference

Git Official Documentation – git-scm.com/doc | Git Pro Book, Ch. 1–2 – git-scm.com/book | Linux man pages – man7.org/linux/man-pages | GNU Coreutils Manual – gnu.org/software/coreutils | Ubuntu/Linux Command Line Basics – <https://ubuntu.com/tutorials/command-line-for-beginners>

The author used Claude to assist in researching and drafting the content and layout of this cheat sheet during preparation. The author reviewed and edited the content as needed and takes full responsibility for the final content of the work.